Multimedia Signal Processing - 2nd module

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1 Question - 10 pts

Consider a gray-scale image with 4 grey levels $L = 0, 1, 2, 3$. The following statistics can be estimated collecting data from an image collection:

$$p(0) = \Pr(l = 0) = 0.25$$
$$p(1) = \Pr(l = 1) = 0.25$$
$$p(2) = \Pr(l = 2) = 0.35$$
$$p(3) = \Pr(l = 2) = 0.15$$

- Compute the entropy of the source.
- Design a Huffman code using block sizes $N = 1, 2$.
- In each case, compute the average code length L_N and generate the compressed bitstream corresponding to the following sequence of pixel values, read in raster scan order: $[1, 1, 2, 0]$.

Inter-pixel correlation is described by the following statistics

$$P = \begin{bmatrix} 0.2 & 0.05 & 0 & 0 \\ 0.05 & 0.2 & 0 & 0 \\ 0 & 0 & 0.3 & 0.05 \\ 0 & 0 & 0.05 & 0.1 \end{bmatrix} \quad (1)$$

where p_{ij} denotes the probability $\Pr(l_1 = i, l_2 = j)$, given two consecutive pixels l_1 and l_2 in raster scan order.

- Show that pixel values of consecutive pixels are statistically dependent
- Compute the joint entropy
- Design a Huffman code using block size $N = 2$ and compute the average code length.

2 Question - 5 pts

Describe the use of short and long term predictors in speech coders.

3 Questions - 5 pts (each answer can be either TRUE or FALSE)

1. Consider a quantization scheme based on compander-expander. Let $y = T(x)$ denote the nonlinear transform and $R_c(D)$ the corresponding rate-distortion function.
   - [T] [F] $T(x) = \int_{-\infty}^{+\infty} p(x)dx$
   - [T] [F] $R_c(D) \geq R_{ECSQ}(D)$, for all $D \geq 0$
   - [T] [F] The output quantization symbols are uniformly distributed.
2. Consider a perceptual audio coder
   - [T] [F] The analysis filter bank is the same used by the perceptual model
   - [T] [F] Noise and tonal maskers are characterized by different SMR
   - [T] [F] The critical bandwidth decreases at higher frequencies
   - [T] [F] The bitrate depends on the audio content to be encoded
3. Consider a hybrid DPCM/DCT video coding system
   - [T] [F] Decoding is computationally more complex than encoding
   - [T] [F] Motion estimation is performed in the DCT domain
   - [T] [F] Temporal redundancy is addressed using DPCM

4 Question - 12 pts

Consider the `rice` image (`I =imread('rice.png');`).
Implement a baseline version of the JPEG compression algorithm with the following simplifications:

- Luminance component only.
- No DPCM coding of DC coefficient.
- No entropy coding.

The quantization matrix is given by (also available in the attached file `Qjpeg.mat`):

$$\begin{bmatrix} 16 & 11 & 10 & 16 & 24 & 40 & 51 & 61 \\ 12 & 12 & 14 & 19 & 26 & 58 & 60 & 55 \\ 14 & 13 & 16 & 24 & 40 & 57 & 69 & 56 \\ 14 & 17 & 22 & 29 & 51 & 87 & 80 & 62 \\ 18 & 22 & 37 & 56 & 68 & 109 & 103 & 77 \\ 24 & 35 & 55 & 64 & 81 & 104 & 113 & 92 \\ 49 & 64 & 78 & 87 & 103 & 121 & 120 & 101 \\ 72 & 92 & 95 & 98 & 112 & 100 & 103 & 99 \end{bmatrix}$$

Compute the rate-distortion curve for the `rice` image by varying the quantization parameter (plot PSNR vs. Q).
Now, consider corrupting the image in the following ways

- Adding white Gaussian noise with standard deviation $\sigma_z = 100$
- Adding a luminance offset

In both cases, compute the rate-distortion curve. Compare the curve with the original one and comment the results.